EECE2140 - Final Project Iteration #1
Group 07 - Drew Bancroft, Kavi Bandara, Anthony Wang

SCOPE

Feature	What it does
Cell/Pack model	Each cell has a capacity, state of charge (SOC), and internal resistance. Terminal voltage comes from finding the open circuit voltage (OCV) at SOC from a table and subtracting the instantaneous voltage drop. The pack voltage is the sum of its cells.
Thermal model	Temperature rises with I^2R heating and cools toward ambient: $T \mathrel{+}= (I^2R - k(T - T_{amb})) * dt / C_{th}$
Charging	A CC-CV charger: constant current until a cell reaches 4.2 V, then constant voltage while the current tapers, then it stops at a cutoff current.
Discharging/Loads	Selectable load profiles: constant current, pulsed, and a profile loaded from a CSV file.
SOC estimation	Coulombs are counted on a "measured" current that includes simulated sensor noise and offset. It is compared against the true SOC to show estimation error.
Energy tracking	Watt-Hours (WH) both in and out, efficiency, and cycle count.
Protection	Over/undervoltage, overcurrent, overtemperature, and low SOC. Each has two levels, WARNING and FAULT. A fault latches and opens a simulated contactor that disconnects the load and charger.
Text UI	A menu to configure and run scenarios, a live status line (time, V, I, SOC as an ASCII bar, temp, active warnings), and an event log.
Logging	Writes every timestep to CSV. You graph it in Excel or Python for the report and slides, which keeps the program text-only.

SELECTED TECH/TOOLS

- This project will be done entirely in C++, implementing the techniques taught throughout the semester. In the same breath, GitHub and Sites will also be employed to document the project and all related components.

TEAM/MEMBER OBJECTIVES

- **Team Objectives:**
  - Agree on a shared architecture/headers before implementing code
  - Maintain organized GitHub repo with feature branches/reviewed PRs
  - Integrate all appropriate modules into one simulation with 4-5 test scenarios
  - Confirm all models agree with hand calculations and record results
  - Write README, presentation, and final report together (split by subsystem)
  - Meet all milestones with definitive feature freeze occurring on 11/22
- **Drew (Control and Safety Lead):**
  - Charging: CC-CV charger to hold constant current until a cell reaches 4.2V, switch to constant voltage, and stop at a set cutoff current
  - SOC estimation: coulomb counting on measured current with sensor noise and offset; report the error against defined SOC
  - Energy tracking: accumulate Wh_{in} and Wh_{out} to compute round-trip efficiency
  - Protection: over/undervoltage, overcurrent, overtemp, and low_SOC checks with established warning and fault levels; fault should disconnect load and charger
  - Testing/Presentation: write tests for CC-CV transition, coulomb counting, fault latching, and present control/safety programs
- **Kavi (Battery/Thermal Modeling Lead):**
  - Cell model: per-cell capacity, SOC, and internal resistance with terminal voltage coming from OCV-SOC lookup table with IR's contribution depending on charge or discharge
  - Pack model: combine cells in series so pack voltage equates to the sum of individual cells
  - Thermal model: implement equation for each cell (have temp feed back into internal resistance)
  - Testing/Presentation: write tests for OCV interpolation and present battery/thermal models
- **Anthony (Simulation/UI Lead):**
  - Discharging/loads: constant current pulsed and CSV-loaded load profiles, including parsing/validating CSV file
  - Simulation loop: provide fixed loop to call each module in right order; integrates other members' code
  - Text UI: build config menu, scenario runner, live status line, and event log
  - Logging: write every timestep to CSV
  - Testing/Presentation: write tests for CSV parser and present demo/results

PROJECT TIMELINE/MILESTONES

- **Milestone 1: Foundation (10/11 - 10/25)**
  - **Code:**
    - Agree on code architecture and write class headers
    - Establish cell parameters and OCV table
    - Implement Cell, Pack, Timestep Loop, Constant-Current Discharge, and CSV logger
  - **Repo**:
    - Set up folder structure and a build system with README skeleton
    - Agree on workflow with feature branches/pull requests (ensure one reviewer/PR)
  - **Presentation:**
    - Start a live doc of decisions/bugs (establish development process)
  - **Finishline:**
    - ./bms runs a 1C discharge runs from full to empty and the voltage curve plotted from CSV appears accurate to a real discharge curve
- **Milestone 2: All Models Completed (10/25 - 11/8)**
  - **Code:**
    - Build thermal model, CC-CV charger, pulsed and CSV load profiles, SOC estimation with sensor noise and offset, and energy tracking with cycle count
    - Check all models against hand calculations to ensure accuracy; for instance, at constant current temperature should settle at $T_{amb} + I^2R/k$
  - **Repo**:
    - Add tests for OCV interpolation, CC-to-CV transition, and coulomb counting; small assert-based tests count
    - Document model parameters and assumptions as they get finalized
  - **Presentation:**
    - Draft slide outline for all required topics
  - **Finishline:**
    - A full charge-discharge cycle produces sensible voltage, temperature and SOC-error plots
- **Milestone 3: UI, Integration, and Feature Freeze (11/8 - 11/22)**
  - **Code:**
    - Implement protection, including thresholds, fault latching, and the contactor disconnecting the charger/load

        - Build the text UI to include configurable menu, live status line with ASCII SOC bar, and an event log
        - Define 4-5 canned scenarios to act as the demo/results section
        - Freeze features by 11/22
    - **Repo**:
        - Write finalized README with an overview, features, build/run instructions, example output, and architecture diagram
    - **Presentation:**
        - Finish a full draft of slides, including architecture diagram
    - **Report:**
        - Outline report and write sections for system design/model equations
    - **Finishline:**
        - All scenarios run end-to-end from the menu, faults trip and latch correctly
- **Milestone 4: Final Format/Ready for Presentation (11/22 - 11/30)**
    - **Code:**
        - Bug fixes only
    - **Repo**:
        - Have teammate clone repo fresh and follow README to ensure complete compliance; tag 1.0 release if successful.
    - **Presentation:**
        - Finalize slides and rehearse presentation and decide between live or recorded demo
    - **Report:**
        - Finish report; write sections on results and future extensions
    - **Finishline:**
        - Presentation is delivered/GitHub repo is tagged and submitted

CONTRIBUTION TABLE

Iteration #1:

- Drew: Project Timeline/Milestones, Team Member Objectives
- Kavi: Scope, Team Member Objectives
- Anthony: Selected Tech/Tools, Team Member Objectives

Iteration #2:

- Drew: Set up repo, makefile, and .gitignore, and draft charger/SocEstimator, and protection headers along with fault thresholds
- Kavi: Choose OCV-SOC table, cell chemistry, and implement cell/pack terminal voltage
- Anthony: Build fixed loop/CSV logger with constant current load so cell can be plotted